

Tribhuvan University
Institute of Science and Technology
2069

Bachelor Level/ First Year/ First Semester/ Science
Computer Science and Information Technology (CSc. 113)
(physics)

Full Marks: 60
Pass Marks: 24
Time: 3 hours.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Set A

Attempt all questions:

1. Discuss the motion of a charged particle in an alternating electric field. (7)
2. What do you mean by non-conservative forces? Also state an explain general law of conservation of energy. (2+2+3)
3. State and explain Gauss's law. Apply it to find the field outside a uniformly charged sphere of radius a . (1+3+3)
4. What do you mean by displacement current? Prove $\nabla \times \vec{H} = \vec{J} + \frac{d\vec{D}}{dt}$. (3.5+3.5)
5. Explain the term power and power factors. Further discuss the phenomena of resonance and hence obtain quality factor. (2+2+2+1)

Set B

Attempt any eight questions:

6. Show the path of one projectile as seen from another projectile will always be a straight line. (4)
7. A rocket is moving upwards with acceleration $3g$. Calculate the effective weight of astronaut sitting in the rocket when his actual weight is 75 Kg. (4)
8. A particle of mass m is moving along a circular path in a plane show that force action on it is conservative. (4)
9. The differential equation for a certain system is $\frac{d^2x}{dt^2} + 2\gamma \frac{dx}{dt} + \omega_0^2 x = 0$. If $\frac{\omega_0}{\gamma} \gg 1$, find the time in which energy of the system falls to $(1/e)$ times the initial value. (4)
10. A water drop is observed to fall through gas of density 0.001 gm/cc with a constant velocity of 980 cm/sec. What is the radius of the drop? (η for the gas = 2×10^{-4} poise) (4)
11. Find the electric field at distance Z above the midpoint of a straight line segment of length $2L$, which carries a uniform line charge λ . (4)
12. Two parallel conducting plates are separated by the distance d and p.d. $\Delta\phi$. A dielectric slab of dielectric constant K is and of uniform thickness is tightly fitted between the plates. Find the electric field in the dielectric. (4)
13. Find the vector potential of an infinite solenoid with N turns per unit length, radius R and current I . (4)

- 14.** The series combination of a resistance R and an inductance L is put in parallel with the series combination of resistance R and capacitance C . Show that if $R^2 = L/C$ the impedance is independent of frequency. (4)
- 15.** Consider a simple RL circuit in which a sudden voltage V is applied. Discuss its transient behavior and find the current as a function of time. (4)